Part 15 Subpart B Engineering Assessment

Supplier's Declaration of Conformity — Faceoff Edge, Model FEC1

1. Product identification

Product	FaceoffEdge shaft-mounted lacrosse training sensor
Model number	FEC1
Equipment class	Class B digital device (unintentional radiator), 47 CFR Part 15 Subpart B
Contains	Certified transmitter module, FCC ID Z4T-XIAONRF52840
Assessment date	August 31, 2026

2. Responsible party

Under 47 CFR §§ 2.906 and 2.909, the responsible party for a Supplier's Declaration of Conformity must be located in the United States.

Responsible party	T & R Technologies, LLC
Address	23 Country Corners Rd, Wayland, MA 01778, USA
Contact	support@faceoffedge.com
Telephone	(617)-784-2337

3. Equipment description

Certified module

The product incorporates a Seeed Studio XIAO nRF52840 module, certified under FCC ID Z4T-XIAONRF52840 for its intentional radiator functions under 47 CFR § 15.247. The module operates Bluetooth Low Energy in the 2.4 GHz ISM band with a 2.0 dBi chip antenna. Transmit power is fixed in firmware at 0 dBm (1 mW).

Host additions

Beyond the certified module, the host product consists entirely of:

- One 3.7 V, 400 mAh lithium-polymer cell connected to the module's battery input pads
- A non-conductive polymer enclosure
- A silicone plug covering the USB-C port opening
- A non-conductive clip integral to the enclosure

Explicitly absent

The host adds no additional printed circuit board, no additional digital circuitry, no additional clock or oscillator source, no switching regulator, no display, no additional transmitter or receiver, and no external cabling or interface ports beyond the module's own USB-C connector.

4. Regulatory basis

A certification granted for a modular transmitter covers the intentional radiator functions of the module only. Under 47 CFR § 15.101(e), the digital circuitry within the module is treated as a subassembly, and FCC KDB Publication 996369 D03 states that the resulting host system must be shown to comply with the appropriate equipment authorization procedure with those subassemblies installed in the host and operating.

The applicable authorization procedure for a Class B digital device is the Supplier's Declaration of Conformity under 47 CFR § 15.101 and § 2.1077. SDoC does not require testing by an accredited laboratory or a Telecommunication Certification Body, and no filing is made with the Commission. The responsible party must retain the compliance record and produce it on request.

Applicable emission limits are those of 47 CFR § 15.109 (radiated) and § 15.107 (conducted) for Class B digital devices. Section 15.107 conducted limits apply to equipment connected to the AC mains; this product is battery powered and is charged from a separate, independently authorized USB power supply not supplied as part of the product.

5. Assessment

Emission sources

The only source of unintentional radiated emissions in this product is the digital circuitry of the certified module — the nRF52840 microcontroller, its onboard oscillators, and its flash memory. This circuitry was present and operating in the configuration the module grantee tested. The host introduces no additional source of emissions.

Effect of the enclosure

The enclosure is a dielectric polymer. It provides no shielding, but it also introduces no conductive structure capable of coupling to, re-radiating, or resonating with emissions from the module. Its effect on the radiated emission profile is therefore expected to be negligible.

Effect of the battery

The lithium-polymer cell is a passive DC source. It contains no switching elements and generates no emissions of its own. Its leads are short relative to a wavelength at the frequencies of interest and are not expected to function as an efficient radiating structure. Charge management is performed by the module's own onboard circuitry, which was included in the module's certification.

Conclusion

Because the host adds no digital circuitry, no new clock sources, and no conductive structures, the unintentional radiated emissions of the composite product are expected to be substantially equivalent to those of the certified module evaluated in isolation. On that basis the product is assessed as compliant with the Class B limits of 47 CFR § 15.109.

6. Declaration

The undersigned has reviewed the construction of the equipment described above and the basis set out in Sections 4 and 5, and declares that the product identified in Section 1 complies with the applicable requirements of 47 CFR Part 15 Subpart B for a Class B digital device.

Signature	
Name	**Trevor Thompson**
Title	**Member**
Qualification	**Product designer and responsible party**

Date 8/31/2026

7. Records retained

The following are held on file by the responsible party and available to the Commission on request:

- This assessment, signed and dated
- FCC grant of certification for module FCC ID Z4T-XIAONRF52840, and the grantee's integration instructions
- Bill of materials for Model FEC1
- Firmware finding recording that Bluetooth transmit power is fixed at 0 dBm with no code path to raise it, supporting the RF exposure exemption under § 1.1307(b)(3)(i)(A)
- UN38.3 test summary for the lithium-polymer cell